

Medical Robotics Project

Control and Trajectory Design for Robotic-Assisted Minimally Invasive Surgeries

1 Goal

The main goal of this project is to implement control and trajectory architectures for minimally invasive surgeries, taking into account geometric, kinematic and dynamic models of planar robots.

2 Control Design

The planar surgical robot represented in Fig. 1 interacts with an eye for delicate surgery. The surgeon grabs directly the tip for operation (comanipulation mode), which should be done along a specific Cartesian direction. Robot dynamics is given by

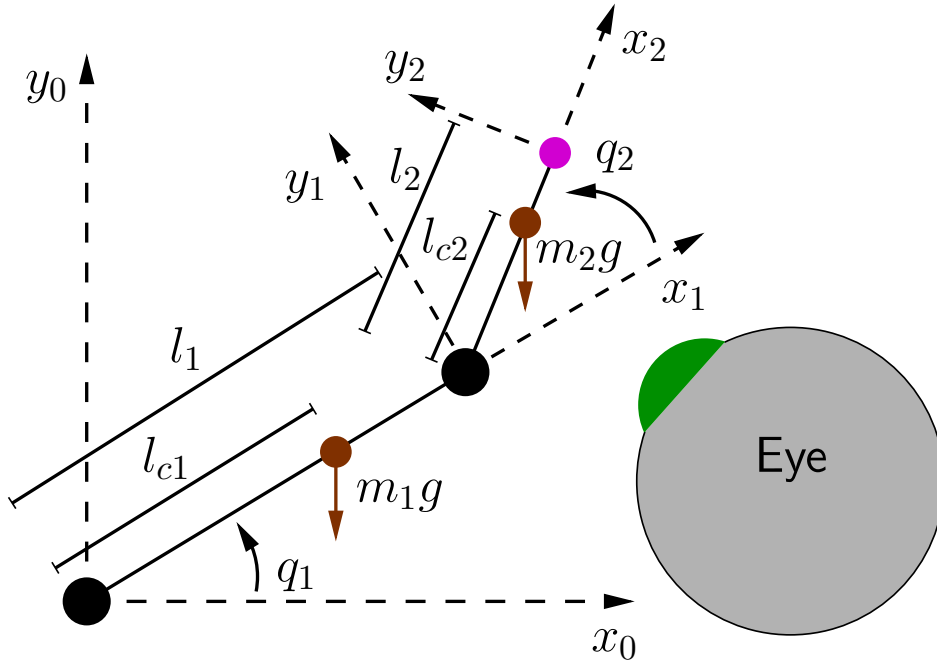


Figure 1: Two-link planar robot for eye surgeries.

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau, \quad (1)$$

where $M(q)$ is the inertia matrix, $C(q, \dot{q})$ represents centripetal and Coriolis terms and $g(q)$ is the gravity term [3]. τ is the generalized torque applied to the joint vector q , i.e.,

$$\tau = \tau_c + \tau_f + \tau_e, \quad (2)$$

where τ_c is the control torque, τ_f is the torque due to friction and τ_e represents all other external torques. Therefore, (1) can be written as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + F_v\dot{q} = \tau_c + \tau_e, \quad (3)$$

where F_v is viscous friction (we are neglecting other friction sources). For this surgical robot consider that $m_1 = 0.3$, $m_2 = 0.1$, $l_1 = 0.3$, $l_2 = 0.2$, $l_{c1} = l_1/2$, $l_{c2} = l_2/2$ and $F_v = 0.04$. The moment of inertia of link i , I_i , about the center of mass is given by $I_i = \frac{1}{12}m_i l_i^2$ along z (all quantities are in SI units).

- 1) Write a Matlab function to compute the forward kinematics. Use explicit mathematical modeling and also AI tools to write the Matlab function. Compare the results.
- 2) Write a Matlab function to compute the Jacobian referred to the end-effector, and the Jacobians referred to each center of mass for both linear and angular motions. Use explicit mathematical modeling and also AI tools to write the Matlab function. Compare the results.
- 3) Relying on the symbolic toolbox, write Matlab functions to compute $M(q)$, $C(q, \dot{q})$ and $g(q)$, based on the Lagrangian formulation. Use explicit mathematical modeling and also AI tools to write the Matlab function. Compare the results.
- 4) Based on Matlab functions and Simulink blocks, obtain a robot simulator. The initial conditions are

$$q_0 = \begin{bmatrix} \frac{\pi}{2} & \frac{\pi}{4} \end{bmatrix}^T \quad (4)$$

and

$$\dot{q}_0 = [0 \ 0]^T. \quad (5)$$

Use graphical functions to visualize the robot. Test the simulator by applying $\tau_c = 0$ and $\tau_e = 0$. What happens to the robot? Play with the simulator for several initial conditions and for different friction terms.

- 5) Apply now $\tau = g(q)$. What happens to the robot?
- 6) Implement a control law that applies Cartesian forces at the robot end-effector with gravity compensation turned on. Apply the Cartesian forces $F_c = [0 \ -10]^T$, $F_c = [10 \ 0]^T$ and $F_c = [10 \ -10]^T$. Analyze the relation between the final robot position and F_c .

- 7) Design, implement and test position control schemes in the task space (P and PD) to control the end-effector position [2], [1]. Generate an elliptical trajectory centered around (0.15,0.2) with major axis $m_x = 0.15$ and minor axis $m_y = 0.1$. Start with a cyclic frequency of 1/3 [Hz].
- 8) Test also diagonal trajectories with arbitrary slope. Analyze the robot behavior for different control gains.
- 9) Design, implement and test compliance control schemes for eye surgery [2], [1]. Make stiff and compliant designs for the robot arm with virtual wall protection. Start the robot in free space, and use a virtual force field for the approaching phase in free space. Simulate surgeon interactions by applying Cartesian forces at the end-effector (2D-forces). Analyze the robot behavior in contact and in free space. Explain design options.
- 10) Design, implement and test an impedance control scheme for eye surgery along a diagonal line constraint [2], [1]. Explain control options.
- 11) Design, implement and test an admittance control scheme for eye surgery [1]. Explain control options.

3 Trajectory Design for Minimally Invasive Abdominal Surgery

Trajectory generation is usually left to the physician for task execution. In case of autonomous motion (e.g., image guided motion), sensory information is mapped into desired task motion. In both cases, robot redundancy can be exploited in multiple ways through null-space motion design. Null-space velocities or null-space torques can be designed to tackle some criterion such as maximizing robot manipulability, avoiding vital organ contacts and keeping a desired posture.

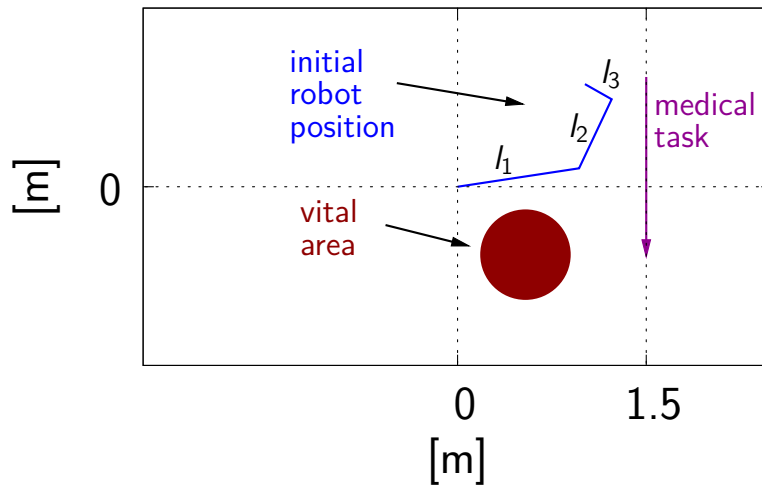


Figure 2: Three-link surgical robot.

- 1) Design, implement and visualize a joint trajectory for the three-link robot represented in Fig. 2, for straight line motion avoiding vital areas. Consider that $l_1 = 1$ [m], $l_2 = 1$ [m] and $l_3 = 0.3$ [m]. Explain design options.
- 2) Design, implement and visualize a joint trajectory for a horizontal motion with an elbow-up posture. Explain design options.

4 Research

- Be creative! Develop your own ideas, and explain them.
- Several lightweight Ufactory robots exist in our labs and can be used to test innovative ideas. Use AI tools to help you out with code design. An image of such robot interacting with a human is represented in Fig. 3.



Figure 3: Lite 6TM robot.

References

- [1] R. Cortesão. Medical robotics course. University of Coimbra, 2020.
- [2] W. Khalil and E. Dombre. *Modeling, Identification and Control of Robots*. Hermes Penton Ltd, 2002.
- [3] M. Spong, S. Hutchinson, and M. Vidyasagar. *Robot Modeling and Control*. John Wiley & Sons, Inc., USA, 2006.